

A sample of $n (> 1)$ units with pps can be drawn with or without replacement. Let us consider a pps-wr sample. Let $\{y_i, p_i\}$ be respectively the sample observation on the selected unit and the initial probability of selection at the i -th draw, $i = 1, 2, \dots, n$. Each (y_i / p_i) , $i = 1, 2, \dots, n$ in the sample is an unbiased estimate $[Y_{i(pps-wr)}^*]$ of the population total Y and $V(Y_{i(pps-wr)}^*) = (Y_r^2 / P_r) - Y^2$, $r = 1$ to n . (see 2.60). Estimates from pps-wr samples are:

$$Y_{pps-wr}^* = (1/n) \sum_{i=1}^n (y_i / p_i) = (1/n) \sum_{i=1}^n Y_{i(pps-wr)}^* ; \quad i = 1 \text{ to } n. \quad (2.62)$$

$$V(Y_{pps-wr}^*) = (1/n) \sum_{r=1}^n (Y_r^2 / P_r) - Y^2 ; \quad r = 1 \text{ to } N. \quad (2.63)$$

$$V(m_{pps-wr}) = (1/N^2) \sum_{r=1}^n (Y_r^2 / P_r) - Y^2 ; \quad r = 1 \text{ to } N. \quad (2.64)$$

$$v(Y_{pps-wr}^*) = [1/\{n(n-1)\}] \sum_{r=1}^n (y_r^2 / p_r^2 - n Y^{*2}) ; \quad r = 1 \text{ to } n; \quad (\text{using 2.51}) \quad (2.65)$$

Operational procedure for drawing a pps-wr sample: The steps are:

- 1) Cumulate the sizes of the units to arrive at the cumulative totals of the unit sizes. Thus

$$T_{i-1} = X_1 + X_2 + \dots + X_{i-1} ; T_i = X_1 + X_2 + \dots + X_{i-1} + X_i = T_{i-1} + X_i ; \quad i = 1, 2, \dots, N.$$

- 2) Then choose a random number R between 1 and $T_N = X_1 + X_2 + \dots + X_N = X$.
- 3) Choose the unit U_i if R lies between T_{i-1} and T_i , that is, if $T_{i-1} < R \leq T_i$. The probability $P(U_i)$ of selecting the i -th unit will thus be $P(U_i) = (T_i - T_{i-1}) / T_N = X_i / X = P_i$.
- 4) Repeat the operation 'n' times for selecting a sample of size n with pps-wr.

7.5.5.6 Stratified Sampling

The Method: We might sometimes find it useful to classify the universe into a number of groups and treat each of these groups as a separate universe for purposes of sampling. Each of these groups is called a **stratum** and the process of grouping **stratification**. Estimates obtained from each stratum can then be combined to arrive at estimates for the entire universe. This method is very useful as (i) it gives estimates not only for the whole universe but also for the sub-universes and (ii) it affords the choice of different sampling methods for different strata as appropriate. It is particularly useful when a survey organisation has regional field offices. This method is called **Stratified Sampling**.

Let us divide the population (universe) of N units into k strata. Let N_s be the number of units in the s -th stratum. Y_{si} be the value of the i -th unit in the s -th stratum. Let the population mean of the s -th stratum be M_s . $M_s = (1/N_s) \sum_{i=1}^{N_s} Y_{si}$, $i = 1, 2, \dots, N_s$ (that is over the units within the s -th stratum) and the population M is $= (1/N) \sum_{s=1}^k N_s M_s = \sum_{s=1}^k W_s M_s$, where $W_s = (N_s / N)$ and $\sum_{s=1}^k W_s = 1$ (being over the strata $s = 1, 2, \dots, k$). Suppose that we select random samples from each stratum and the sampling method for different strata are different. Let the unbiased estimate of the population mean M_s of the s -th stratum be m_s . Denoting 'st' for stratified sampling, **an unbiased estimator of M is given by**

$$m_{st} = \sum_{s=1}^k W_s m_s = (1/N) \sum_{s=1}^k N_s m_s, \quad s = 1 \text{ to } k. \quad (2.66)$$

$$V(m_{st}) = \sum_{s=1}^k W_s^2 V(m_s) = (1/N^2) \sum_{s=1}^k N_s^2 V(m_s), \quad s = 1 \text{ to } k \quad (2.67)$$

$$\text{Cov.}(m_s, m_r) = 0 \text{ for } s \neq r ; (\text{samples from diff. strata are independently chosen}) \dots \quad (2.68)$$

$$Y_{st}^* = \sum_s Y_s^* ; \quad s, s = 1 \text{ to } k. \quad (2.69)$$

$$V(Y_{st}^*) = \sum_s V(Y_s^*), \quad s, s = 1 \text{ to } k. \quad (2.70)$$

Thus estimators with smaller variance (efficient estimators) can be obtained in stratified sampling if we form the strata in such a way as to minimise *intra-strata* or within-strata variation, that is, variance within strata. This would mean maximising between-strata or *inter-strata* variation, since the total variation is made up of within-strata and between-strata variation. In other words, *units in a stratum should be homogeneous*.

Stratified sampling enables us to choose the sample we wish to select by drawing independent samples from each of the different strata in to which we have grouped the universe. *How do we allocate the total sample size 'n' among the different strata?* One way is to *allocate the sample size to different strata in proportion to the size of individual strata measured by the number of units in these strata, namely, N_s , [$\sum_s N_s = N$, ($s = 1 \text{ to } k$)]*. This method is especially appropriate in situations where no information is available except the sizes of the strata. The sample sizes for the samples from the stratum, say, the s -th stratum, would then be $n_s = n(N_s/N)$ and $\sum_s n_s$ can easily seen to be equal to 'n'. There are other methods like allocation of the sample size among strata in proportion to the stratum totals of the variable under study, that is, Y_s , the stratum total of the s -th stratum, ' $s = 1 \text{ to } k$ '. We shall not go into the details of other methods here except one situation, namely, ***when we have a fixed budget F sanctioned for the survey***. Let the cost function F be of the form $F_0 + \sum_s n_s F_s$, ($s, s = 1 \text{ to } k$), where F_0 , n_s and F_s are respectively the overhead cost, the sample size in stratum ' s ' and the per unit cost of surveying a unit in stratum ' s ' ($s = 1, 2, \dots, k$). We can determine the optimum stratum-wise-sample-size by minimising the sampling variance of the sample mean (2.67) subject to the constraint that the cost of the survey is fixed. The stratum-wise optimum sample size is given by

$$n_s = [(F - F_0)] [W_s \sqrt{(V_s / F_s)}] / [\sum_s W_s \sqrt{(V_s F_s)}], \quad s = 1, k. \quad (2.71)$$

The stratum sample size should, therefore, be proportional to $W_s \sqrt{(V_s / F_s)}$. The minimum variance with the n_s so determined is,

$$\text{Min. } V(m_{st}) = [\sum W_s \sqrt{(V_s F_s)}]^2 / (F - F_0) \quad (2.72)$$

Check Your Progress 4

1) When is PPS method adopted?

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2) When will the sampling variance of Y_{pps}^* be small?

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3) Say True (T) and False (F):

- PPS and stratified sampling can be combined with other sampling methods. (T/F)
- $V(m_{st})$ is reduced by ensuring that units within individual strata are heterogeneous. (T/F)
- The size of a stratified sample can be allocated among the strata, the size being the number of population units in a stratum. (T/F)

4) A systematic sample of size 18 has to be selected from a population of 124. What problems do you face in selecting the sample? Is the sample mean the unbiased estimator of the population mean M ? How do you overcome these problems?

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7.5.4.7 Cluster Sampling

The method: Supposing we are interested in studying certain characteristics of individuals in an area. We would naturally select a random sample of individuals from all the individuals residing in the area and collect the required information from the selected individuals. We might also think of selecting a sample of households out of all the households in the area and collect the required details from all the individuals in the selected households. *The households in the area are clusters of individuals and what we have done is to select a sample of such clusters and to collect the information needed from all the individuals in the selected clusters instead of selecting a random sample of individuals from all persons in the area.* What we have done is **cluster sampling**. **Cluster Sampling is a process of forming suitable clusters of units and surveying all the units in a sample of clusters selected according to an appropriate sampling method. The clusters of units are formed by grouping neighbouring units or units that can be conveniently surveyed together.** Sampling methods like srswr, srswor, systematic sampling, pps and stratified sampling discussed earlier can be applied to sampling of clusters by treating clusters themselves as sampling units. The clusters can all be of equal size or varying sizes, that is, the number of units can be the same, or vary from cluster to cluster. Clusters can be mutually exclusive, that is, a unit belonging to one cluster will not belong to any other cluster. They could also be overlapping.

Estimates from cluster sampling: Let us consider a population of NK units divided into N mutually exclusive clusters of K units each – a case of clusters of equal size. The population mean M and the cluster means are given respectively by $M = (1/N) \sum_{s=1}^N m_s$, m_s being over clusters $s = 1$ to N and $m_s = (1/K) \sum_{i=1}^K Y_{si}$, Y_{si} being from $i = 1$ to K within the s -th cluster. Let us draw a sample of one cluster by srs. **The cluster mean m_{c-srs} (the subscript $c-srs$ denotes cluster sampling with srs) is an unbiased estimate of M .** The sampling variance of the sample cluster mean is

$$V(m_{c-srs}) = (1/N) \sum_{s=1}^N (m_s - M)^2 = \sigma_b^2 = \text{Variance between clusters}; \quad s = 1 \text{ to } N; \quad (2.73)$$

Let us compare $V(m_{c-srs})$ with the sampling variance of the sample mean *when K units are drawn from NK units by SRSWR method*. How does the “sampling efficiency” of cluster sampling compare with that of SRSWR? ***The sampling efficiency of cluster sampling compared to that of SRSWR, $E_{c/srswr}$, is defined as the ratio of the reciprocals of the sampling variances of the unbiased estimators of the population mean obtained from the two sampling methods.*** The sampling variances and sampling efficiency are

$$V(m_{srswr}) = (1/K) [(1/NK) \sum_{i=1}^N Y_{si}^2 - M^2] = \sigma^2/K \quad (2.74)$$

$$\sigma^2 = \sigma_w^2 + \sigma_b^2 = \text{within-cluster variance} + \text{between-cluster variance.} \quad (2.75)$$

$$E_{c/srswr} > 1 \text{ if } \sigma_w^2 > (K-1)\sigma_b^2 \quad (2.76)$$

Thus, cluster sampling is more efficient than SRSWR if the *within-cluster* variance is larger than $(K-1)$ times the *between-cluster* variance. Is this likely? This is not likely as the *between-cluster* variance will usually be larger than the *within-cluster* variance due to *within-cluster* homogeneity. ***Cluster sampling is in general less efficient than sampling of individual units from the point of view of sampling variance.*** Sampling of individuals could provide a better cross section of the population than a sample of clusters since units in a cluster tend to be similar.

7.5.4.8 Multi-Stage Sampling

We noted in the sub-section on cluster sampling that random sampling of units directly is more efficient than random sampling of clusters of units. But cluster sampling is operationally convenient. How to get over this dilemma? We may first select a random sample of clusters of units and thereafter select a random sample of individual units from the selected clusters. We are thus selecting a sample of units, but from selected clusters of units. What we are attempting is a ***two-stage sampling***. This can thus be a compromise between the efficiency of direct sampling of units and the relatively less efficient sampling of clusters of units. This type of sampling would be more efficient than cluster sampling but less efficient than direct sampling of individual units. In the sampling procedure now proposed, the clusters of units are ***the first stage units (fsu)*** or ***the primary stage units (psu)***. The individual units constitute ***the second stage units (ssu)*** or ***the ultimate stage units (usu)***.

This procedure of sampling can also be generalised to multi-stage sampling. Take for instance a rural household survey. The fsu's in such a survey may consist of districts, the ssu's may be the *tehsils or taluks* chosen from the districts selected in the first stage, the third stage units could be the villages selected from the *tehsils or taluks* selected in the second stage and the fourth and the ultimate stage units (usu's) may be the households selected from the villages selected in the third stage. Such multi-stage sampling procedures help in utilising such information related to the variable under study as may be available in choosing the sampling method appropriate at different stages of sampling. In a multi-stage sampling, estimates of parameters are built up stage by stage. For instance, in two-stage sampling, estimates of the *sample aggregates relating to the fsu's* are built up from the ssu's using the sampling method adopted for selecting the ssu's. These estimates are then used with the sample probabilities of selection of fsu's to build up estimates of the relevant population parameters.

7.6 THE CHOICE OF AN APPROPRIATE SAMPLING METHOD

We have considered a number of random sampling methods in the foregoing sub-sections. A natural question that arises now is – *which method is to be adopted in a given situation?* Let us consider this question, although the answer to it lies scattered across the foregoing sub-sections. The choice of a sampling design depends on considerations like *a priori* information available about the population, the precision of the estimates that a sampling design can give, operational convenience and cost considerations.

- 1) When we do not have any *a priori* information about the nature of the population variable under study, SRSWR and SRSWOR would be appropriate. Both are operationally simple. However, ***SRSWOR is to be preferred***, since $V(m_{\text{srswor}}) < V(m_{\text{srswr}})$. This advantage holds *only when the sampling fraction is not small, or N and n are not large*.
- 2) Systematic sampling is operationally even simpler than SRSWR and SRSWOR, but it should not be used for sampling from populations where periodic or cyclic trends/variations exist, though this difficulty can be overcome if the period of the cycle is known. $V(m_{\text{sys}})$ can be reduced if the units chosen in the sample are *as heterogeneous as possible*. But this will call for a rearrangement of the population units before sampling.
- 3) When additional information is available about the variable ‘y’ under study, say, on a variable (size variable) ‘x’ related to ‘y’, the pps method should be preferred. The sampling variance of Y^* (or m) gets reduced when the probability of selection of units $P_i = (X_i / N)$ are proportional to Y_i , that is, the size X_i is proportional to Y_i or the variables x and y are linearly related to each other and the regression line passes through the origin. In such cases pps is more efficient than SRSWR. Further, this method can be utilised along with other sampling methods and their relative efficiencies. *pps is operationally simple*. Pps-wor combines the efficiency of SRSWOR and the efficiency-enhancing capacity of pps. However, most of the procedures of selection available, estimators and their variance for pps-wor are complicated and are not commonly used in practice. This is particularly so in large-scale sample surveys with a small sampling fraction, as in such cases sampling without replacement does *not* result in much gain in efficiency. Hence unless the sample size is small, we should prefer pps-wr.
- 4) Stratified sampling comes in handy when we wish to get estimates at the level of sub-populations or regions or groups. This method also gives us the freedom to choose different sampling methods/designs in different strata as appropriate to the group (stratum) of the population and the opportunity to utilise available additional information relating to the stratum. The sampling variance of estimators can also be brought down by forming the strata in such a way as to ensure *homogeneity of units within individual strata*. In fact, the stratum sizes can be so chosen as to minimise the variance of estimators, when there is a ceiling on the cost of the survey. ***Stratified sampling with SRS, SRSWOR or pps-wr presents a set of efficient sampling designs.***
- 5) Sometimes, sampling of groups of individual units than direct sampling of units might be found to be operationally convenient. Supposing it is easier to get a complete frame of clusters of individual units than that of units or,

only such a frame, and not that of the units, is available. (e.g. households are clusters of individuals). In such circumstances, cluster sampling is adopted. *This is in general less efficient than direct sampling of individual units, as clusters usually consist of homogeneous units. A compromise between operational convenience and efficiency could be made by adopting a two-stage sampling design*, by selecting a sample of individual units (second stage units) from sampled clusters (the first stage units). A multi-stage design would be useful in cases where clusters have to be selected at more than one stage of sampling.

- 6) Finally, we can use the technique of independent I-PSS in conjunction with the chosen sampling design to get at (i) an unbiased estimate of $V(m)$ for any sampling design or estimator of $V(m)$, however complicated, (ii) a confidence interval for 'M' based only on the I-PSS estimates (when the population distribution is symmetrical) and (iii) a tool for monitoring the quality of work of the field staff and agencies.
- 7) *SRSWOR, stratified sampling with SRSWOR and, when available information permits, pps-wr and stratified sampling with pps-wr, turn out to be a set of the more efficient and operationally easy designs to choose from. I-PSS can also be used in these designs where possible and necessary.*

Check Your Progress 5

- 1) We wish to study the wage levels of factory labour. What type of sampling method would you adopt for the study and why if (a) just a list of factories is available with the Chief Inspector of Factories of different State Governments, (b) if the list in (a) above also gives the total number of employees in the individual factories at the end of last year and (c) the list also indicates both the kind of product manufactured in the factory along with the information specified in (b) above.

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7.7 LET US SUM UP

There are broadly three methods of collecting data. The array of tools used for data collection by such methods has expanded over time with the advent of modern technology. Confining data collection efforts to a sample from the population of interest to the study inevitably leads to questions like the use of random and non-random samples. Judgment sampling, convenience sampling, purposive sampling, quota sampling and snowball sampling all belong to the latter group. The absence of a clear relationship between a non-random sample and the parent universe and the presence of the researcher's bias in the selection of the sample render such samples useless for drawing *valid* conclusions about the parent population. But these methods are inexpensive and quick ways of getting a preliminary idea of the universe for use in designing a detailed enquiry and in exploratory research. Random samples, on the other hand, are free from such drawbacks and have properties that help in arriving at valid conclusions about the parent population.

The simplest of the sampling methods – **SRSWR** - ensures equal chance of selection to every unit of the population and yields a sample in which one or more units may occur more than once. ' m_{srswr} ' is an unbiased estimator of M . Its precision as an estimator of M increases as the sample size increases. **SRSWOR** yields a sample of *distinct* units. ' m_{srswor} ' is also unbiased for ' M '. ***SRSWOR is a more efficient than SRSWR as $V(m_{\text{srswor}}) < V(m_{\text{srswr}})$. But this advantage disappears when the sampling fraction is small (< 0.05).*** Both provide an unbiased estimator of $V(m)$. An operationally convenient procedure - ***interpenetrating sub-samples (I-PSS)*** – also provides an unbiased estimator of $V(m)$ for any sampling design and estimator for $V(m)$, however complicated.

Systematic sampling is a simple and operationally convenient method used in large-scale surveys that requires only a random start and the sampling interval $k = [N/n]$ for drawing the sample. A slight variant of ' m ' is unbiased for ' M '. ***Circular systematic sampling*** takes care of problems that arise when N/n is not an integer. An unbiased estimate of $V(m)$ is not possible but this problem can be tackled easily. ***Systematic sampling is not recommended when there is a periodic or cyclic variation in the population. This problem too can be overcome if the period of the cycle is known.***

An example of methods where the probability of selection varies from unit to unit is **pps**. The “size” could be the value of a variable related to the study variable. In pps, each y_i / p_i , where y_i is the value of the study variable associated with the selected unit and p_i the probability of selection of the unit, is an unbiased estimate (Y^*) of the population total Y and $[(1/N) \sum Y^*]$ an unbiased estimator of M . As $V(Y^*)$ is small if the probabilities P_i are roughly proportional to Y_i , ***pps sampling is more efficient than SRS if the size variable x is proportional to y , that is, x and y are linearly related and the regression line passes through the origin.*** pps sampling can be done with SRSWR, SRSWOR or systematic sampling. In pps-srswr, $[(1/n) \sum (y_i / p_i)]$, ($i = 1$ to n), is an unbiased estimator of Y . This being the mean of n independent unbiased estimates with the same variance $V(Y^*)$, $v(Y^*)$ can be derived using the I-PSS technique.

Stratified Sampling is used when (i) estimates are needed for subgroups of a universe or (ii) the subgroups could be treated as sub-universes. It gives us the freedom to choose the sampling method as appropriate to each stratum. Estimates of parameters are available for the sub-universes (strata) and these can then be combined over the strata to get estimates for the entire universe. ***SE of estimates based on stratified sampling can be small if we form the strata in such a way as to minimise intra-strata variance. Each stratum should thus consist of homogeneous units, as far as possible. Stratum-wise sample sizes can also so chosen as to minimise the variance of estimators.***

Another operationally convenient sampling method, **cluster sampling**, is to sample groups of units or clusters of units at random and collect data from *all* the units of the selected clusters. For example, the household is a cluster of individuals. SRSWR, SRSWOR, pps or systematic sampling can be used for sampling clusters. ***Cluster sampling is, in general, less efficient than direct sampling of units from the point of view of sampling variance. The question here is one of striking a balance between operational convenience and cost reduction on the one hand and efficiency of the sampling design on the other.***

We could improve the efficiency of cluster sampling by selecting a random sample of units from each of the selected clusters - introduce another stage of

sampling. This is **two-stage sampling**. *This would be more efficient than cluster sampling but less efficient than direct sampling of units.* **Multi-stage sampling** can also be done. Such designs are commonly used in large-scale surveys as these facilitate the utilisation of information available and the choice of appropriate sampling designs at different stages.

Thus while non-random sampling methods are useful in exploratory research and preliminary work on planning of enquiries, random sampling techniques lead to *valid* judgments regarding the universe. Among random sampling methods, **SRSWOR, stratified sampling with SRSWOR and, when available information permits, pps-wr and stratified sampling with pps-wr, turn out to be a set of the more efficient and practically useful designs to choose from. I-PSS can also be used in these designs where possible and necessary.**

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7.9 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Three methods of data collection are: the census and survey method, the observation method, and the experimental method.
- 2) A function of population values is a parameter; a function of sample observations is a statistic.
$$M = \frac{1}{N} \sum Y_i, m = \frac{1}{N} \sum y_i, \sigma^2 = \frac{1}{N} \sum Y_i^2 - M^2, \text{ and } s^2 = \frac{1}{N} \sum y_i^2 - m^2,$$
where M/m is the mean of the population/sample values are the expressions of population/sample means and population/sample variance.
- 3) The purpose of drawing/studying a sample is to arrive at inferences about the characteristics of the unknown population. Thus, the sample mean is an estimator of the population mean (of the variable under study). The actual value of the estimator when computed from a sample is the estimate.
- 4) A representative sample is one that possesses the characteristics of the population in the same proportion as in the population. A conscious or subjective selection of units introduces a bias that erodes the sample's capacity to be a true representative of the characteristics of the population. It is for this reason that a random sampling procedure is preferred. Although non-random sampling is drawn without any regard to the rigors of a random sampling procedure, it gives a rough idea of the unknown features of the population under focus. Implemented judiciously, it can serve the objectives of providing preliminary inputs for planning a detailed enquiry later.

Check Your Progress 2

- 1) Sample observations in a random sample are random variables. A statistic is a function of sample observations and is, therefore, also a random variable. This random variable assumes different values in repeated random samples. These values form a frequency distribution of the statistic. This frequency distribution will, in the limit, tend to a probability distribution that gives the probabilities of the statistic assuming different values. This probability distribution is called the sampling distribution of the statistic. Let a statistic 't' be an estimator of a population parameter, say, T. The estimator 't' is an unbiased estimator of the parameter 'T', if the mean of the sampling distribution of 't' is 'T'. Since the mean of the sampling distribution of 't' can also be denoted by $E(t)$, we can express this condition as $E(t) = T$.
- 2) Let 't' be an estimator of 'T'. The sampling distribution of the statistic 't' can be derived *only* if the statistic is based on a random sample selected from the population with a parameter 'T'. The mean of the sampling distribution can then be derived from the sampling distribution of 't'. That is, $E(t)$ can be derived. If $E(t) \neq T$ and $E(t) = KT$ or $T+K$, we can correct the estimator 't' as (t/K) or $(t - K)$ to arrive at an unbiased estimator of T. An example is the unbiased estimator of σ^2 .
- 3) See Sub-section 7.5.3 A

Check Your Progress 3

- 1) The answer for (i) would be $\sqrt{\frac{4.41}{16}} = \frac{2.1}{4} = 0.525$. The answer for (ii) is $\sqrt{\frac{4.41}{25}} = \frac{2.1}{5} = 0.42$. The percentage decrease in SE works out to $[(0.525 - 0.42) \times 100] / [0.525] = 20\%$. The percentage increase in the sample size is $\frac{9}{16} \times 100 = 56.25\%$. The extent of increase is larger than the extent of decrease in the standard error. Try to work out (iii) and compare the results of (ii) & (iii).
- 2) Use the formula 2.39 for *variance* of the sample mean under SRSWOR. Note that, *for the same sample size*,

$$V(m_{SRSWOR}) - \frac{N-n}{N-1} V(m_{SRSWR}) = 1 - \frac{n-1}{N-1} V(m_{SRSWR}).$$

For $n=16$ and using SRSWOR sampling, SE works out to $\frac{64}{79} \times 0.525 = 0.425$. Compare it with the SE for $n = 25$ in problem 1. SE of the sample mean was reduced to more or less this level by *increasing the size of the SRSWR sample from 16 to 25*. Work out (ii) & (iii)
- 3) a) F. SE(m) decreases in inverse proportion to the square root of the sample size.
 b) F. SRSWOR is more advantageous if the sampling fraction is > 0.5 .
 c) F. $V(Y^*) = N^2 V(m)$.

Check Your Progress 4

- 1) When information on an auxiliary variable 'x' related to the variable 'y' under study is available for each unit of the population.
- 2) When P_i are proportional to Y_i . That is, when the variable 'x' and 'y' are linearly related to each other and the regression line passes through the origin.
- 3) (a) T; (b) F: they should be homogeneous; (c) T.
- 4) The integer 'k' nearest to $124/18$ is 7. Choose a random start between 1 and 7 for selecting the sample. Since $N = 124$ is not a multiple of $n = 18$, two problems arise. (i) Samples with random start 1 to 5 are of size 18 and others of size 17. The statistic $\frac{k}{N} \sum y_i = \frac{7}{124} \sum y_i$ and *not* the sample mean is an unbiased estimate of M. Both the problems at (i) and (ii) are overcome by adopting the circular systematic sampling procedure.

Check Your Progress 5

- 1) a) State-wise lists are available. Stratified sampling with each State (or part of it depending on the number of factories) being a stratum is an obvious choice. We need to collect details relating to *workers* in the factories. Within each stratum a cluster of factories can be selected and wage data relating to *all the workers* in each selected factory may be collected. Alternatively, we could adopt a two-stage sampling – select a sample of factories (fsu's) and a sample of workers (ssu's) from the selected factories, to enhance the efficiency of sampling. SRSWOR would be appropriate for each stage of selection.